

	Case Name: SCM System	Sector	IT
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	Mieke De Saedeleer		User defined distributions
Date	2016	Project Control	Automatic tracking
File Name	C2016-08 SCM System		Tracking based on user input

1. Project description

Project authenticity

SCM System is a project focused on the analysis, integration, and full implementation of supply chain planning processes, including order allocation, DRP, Master Planning, and S&OP. It progresses through validation, system integration, and phased go-lives across multiple plants and markets, delivering a fully operational and optimized planning system.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	124	Serial/Parallel (SP)	49%
Planned Duration (PD)	725d	Activity Distribution (AD)	59%
Budget At Completion (BAC)	375,253.34 €	Length of Arcs (LA)	8%
Renewable Resources	-	Topological Float (TF)	52%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	9	9	1.38
CRI-rho	20	18	0.83
CRI-tau	28	40	1.24

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	32	36	0.77
SI	30	34	0.70
SSI	2	7	7.06
CRI-r	8	10	3.21
CRI-rho	19	19	0.98
CRI-tau	28	40	1.23

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	3.26%	2.43%	1	1.89%	0.20%
PV - SPI	7.66%	7.19%	CPI	4.62%	4.61%
PV - SCI	12.67%	-7.40%	SPI	1.89%	0.20%
ED - 1	2.64%	1.29%	SPI(t)	8.67%	8.66%
ED - SPI	7.63%	7.16%	SCI	9.18%	9.18%
ED - SCI	6.06%	0.75%	SCI(t)	5.25%	4.74%
ES - 1	3.70%	3.10%	0.8 CPI + 0.2 SPI	5.68%	5.37%
ES - SPI(t)	10.08%	9.91%	0.8 CPI + 0.2 SPI(t)	2.91%	2.51%
ES - SCI(t)	6.02%	3.97%			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 34 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-55166.34	-33023.32	-72.27	0.77	0.80	0.73	0.97
std dev	21819.04	23136.83	60.55	0.07	0.17	0.18	0.06
final	-63488.31	0.00	-15	0.86	1.00	0.98	1.00

2.3.3.2. Time forecasting

PD	725d	Real Duration	725d	Late/early	0%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	914.15	44.74	26.09%	-26.09%
PV - SPI	1098.62	319.76	51.53%	-51.53%
PV - SCI	1408.47	498.38	94.27%	-94.27%
ED - 1	895.74	34.56	23.55%	-23.55%
ED - SPI	1099.06	319.41	51.59%	-51.59%
ED - SCI	1305.97	524.37	80.13%	-80.13%
ES - 1	922.71	60.70	27.27%	-27.27%
ES - SPI(t)	1182.21	252.58	63.06%	-63.06%
ES - SCI(t)	1416.03	426.31	95.31%	-95.31%

2.3.3.3. Cost forecasting

BAC	375,253.34 €	Real Cost	438,741.66€	Over/under budget	17%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	430419.68	21819.04	3.49%	1.90%
CPI	488811.31	44039.07	12.26%	-11.41%
SPI	525883.48	144829.15	19.86%	-19.86%
SPI(t)	540698.16	99273.18	23.24%	-23.24%
SCI	619113.54	246596.11	41.11%	-41.11%
SCI(t)	637730.90	179404.38	45.35%	-45.35%
0.8 CPI + 0.2 SPI	492493.88	48272.30	12.53%	-12.25%
0.8 CPI + 0.2 SPI(t)	496495.41	47874.22	13.40%	-13.16%